# Heart ECG Monitor

by ohmies

User Manual

## 1. Introduction

The Electrocardiogram (ECG) Machine is an electronic medical device designed to acquire, amplify, and display the electrical activity of the heart. By detecting the bioelectrical signals generated during each cardiac cycle, the device produces an ECG waveform that represents heart rhythm and electrical conduction.

This user manual provides comprehensive instructions for the installation, operation, and basic care of the ECG machine. It is intended to assist users in understanding the device functionality, ensuring correct usage, and maintaining reliable performance. The manual also outlines important safety information to help prevent improper operation and potential hazards.

The ECG machine enables monitoring of fundamental ECG features such as the P wave, QRS complex, and T wave, which are essential for observing cardiac activity. Proper adherence to the procedures described in this manual ensures accurate signal acquisition and stable operation of the device.

Users are advised to read this manual carefully before operating the ECG machine and to follow all instructions and warnings provided to ensure safe and effective use.

## 2. Package Contents

- ECG Monitor unit
- ECG input cable with 3 inputs (LA, RA, RL)
- 6 ECG electrodes
- Charging module
- User manual

# 3. Safety Information

To ensure safe operation and prevent potential hazards, carefully observe the following safety instructions when using the ECG monitor. Failure to follow these instructions may result in equipment damage, inaccurate readings, or personal injury.

## 0.1 General Safety Warnings

- This device is not intended for medical diagnosis, treatment, or clinical decision-making.
- The ECG monitor is designed for external, non-invasive signal acquisition only.
- Use the device strictly according to the instructions provided in this user manual.
- Do not modify the device circuitry, enclosure, or firmware. Unauthorized modifications may compromise safety and performance.

## 0.2 Electrical Safety

- The ECG monitor is powered exclusively by rechargeable Li-Po batteries.
- Do not connect the device to any external power source other than the specified charging method.
- Use only the recommended charger and charging circuitry.
- Do not operate the device while charging unless explicitly specified.
- Do not short-circuit, puncture, overheat, or disassemble the batteries.
- Stop using the device immediately if the battery shows signs of swelling, leakage, excessive heating, or odor.
- Recharge batteries only within the specified voltage and current limits.

## 0.3 Patient Safety

- Do not use the device on individuals with implanted electronic devices, such as pacemakers or implantable cardioverter-defibrillators (ICDs).
- Use only insulated, skin-safe ECG electrodes intended for bio-signal acquisition.
- Ensure correct electrode placement before starting the measurement.
- Do not place electrodes on broken, irritated, or infected skin.
- Discontinue use immediately if the subject experiences discomfort, pain, or skin irritation.

## 0.4 Environmental Conditions

- Keep the device away from moisture, water, and liquids.
- Do not operate the ECG monitor in wet, humid, or explosive environments.
- Avoid exposure to extreme temperatures, direct sunlight, or excessive dust.
- Operate the device only within the specified environmental conditions.

## 0.5 Handling and Maintenance

- Handle the ECG monitor with care to prevent mechanical damage.
- Do not drop, crush, or apply excessive force to the device or electrodes.
- Clean the enclosure using a soft, dry cloth. Do not use solvents or abrasive cleaning agents.
- Disconnect electrodes from the subject before cleaning or storing the device.

## 0.6 Electromagnetic Interference

- Keep the ECG monitor away from strong electromagnetic sources such as mobile phones, power inverters, motors, and RF transmitters.
- Electromagnetic interference may cause waveform distortion or unstable readings.

## 0.7 Disposal

- Store the device in a clean, dry location when not in use.
- Dispose of the device, batteries, and accessories in accordance with local regulations for electronic and battery waste.
- Do not dispose of batteries in fire or household waste.

# 4. Device Overview

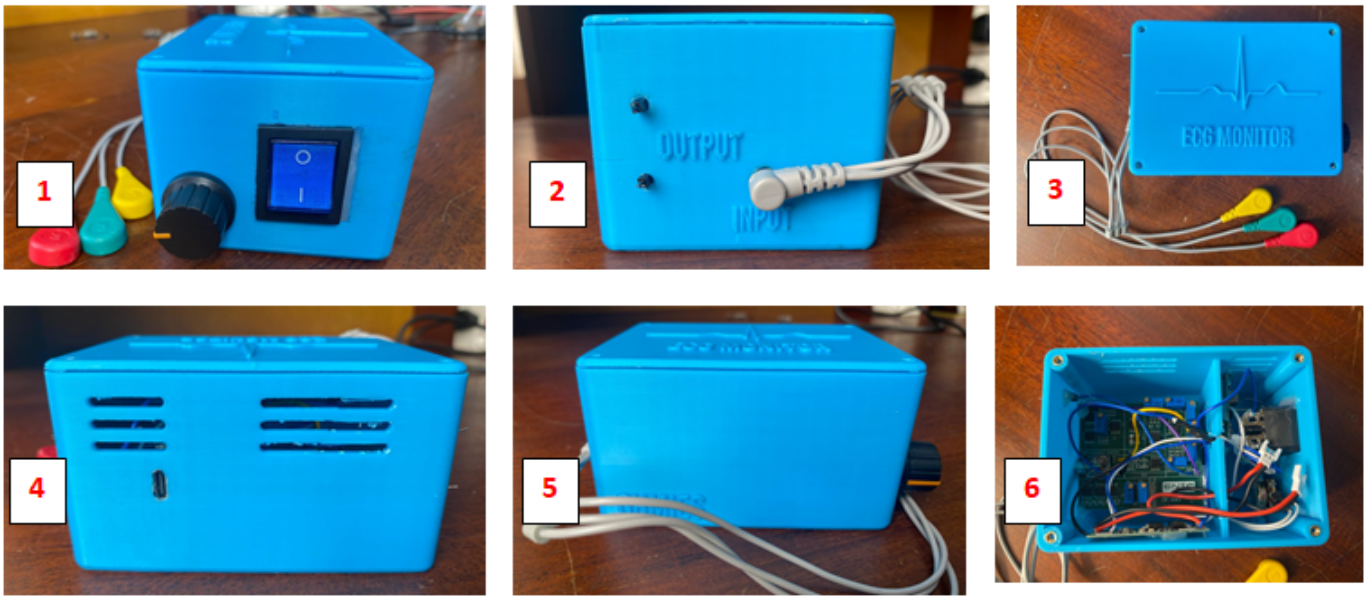



Figure 1 : Power switch and Amplitude controller
Figure 2 : Input port and output pins
Figure 3 : Top View
Figure 4 : Rear View
Figure 5 : Front View
Figure 6 : Inside View with circuit and battery compartment

# 5. Controls and Indicators

- **Power Input:** 5V DC (charging)
- **Battery Voltage:** 3.7v Li-Po battery x 2
- **Electrode Ports:** LA, RA, RL
- **Analog Output:** ECG waveform output

# 6. How to Connect the Electrodes

1. Ensure the device is powered OFF before connecting the electrodes.
2. Clean the electrode placement areas with alcohol and allow the skin to dry completely.
3. Attach the RA (Red) (Right Arm) electrode to the right arm or right shoulder area.
4. Attach the LA (Green) (Left Arm) electrode to the left arm or left shoulder area.
5. Attach the RL (yellow) (Right Leg / Reference) electrode to the lower abdomen or waist area.
6. Verify that all electrodes are firmly attached to the skin.
7. Connect the electrode lead wires securely to their corresponding connectors on the device.
8. connect the output pins to the display through cables
9. Power ON the device and confirm that a stable ECG waveform is displayed and tune the waveform if the signal is unclear by using the Amplitude controller.

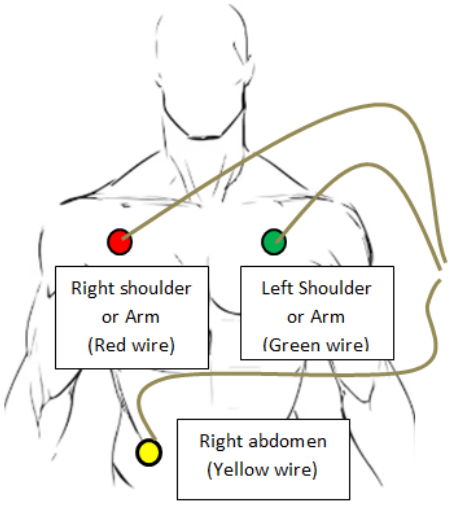



figure : ECG electrode connection

**Note**

- Ensure good skin contact to reduce motion artifacts and noise.
- Avoid placing electrodes over bone, muscle tension areas, or excessive hair.
- Reposition electrodes if the signal appears unstable or noisy.

**Caution**

- Do not attach electrodes to broken or irritated skin.
- Disconnect electrodes immediately after use.

# 7. Operating Instructions

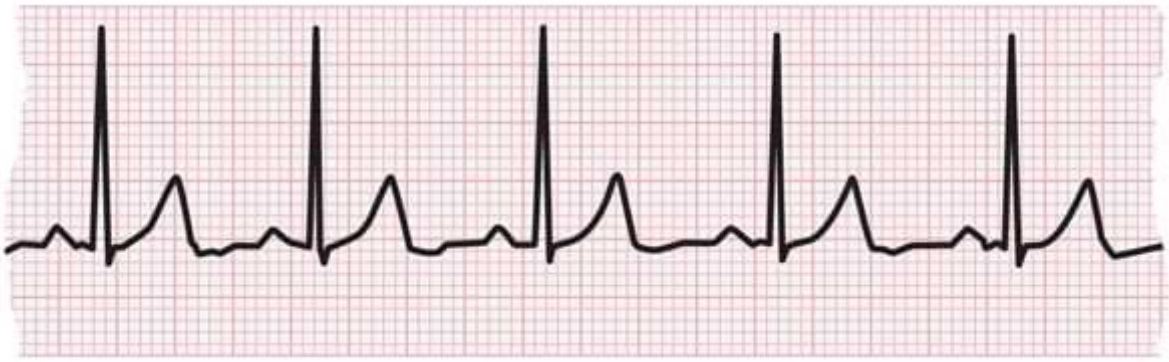
Follow the steps below to operate the ECG monitor correctly.

1. Ensure the device batteries are sufficiently charged.
2. Confirm that the electrodes are connected to the subject as described in Section 6.
3. Connect the ECG output to the monitoring interface (oscilloscope or ADC input), if applicable.
4. Power ON the device using the power switch.
5. Allow a few seconds for the signal to stabilize.
6. Observe the ECG waveform, including the P wave, QRS complex, and T wave
7. Tune the output waveform by adjusting the amplitude controller.
8. Minimize subject movement during measurement to reduce motion artifacts.
9. After use, power OFF the device and disconnect the electrodes from the subject.

**Note**

- Initial signal instability may occur during startup.
- Proper grounding and reference electrode placement improve signal quality.

# 8. Expected Output



# 9.Troubleshooting

This section provides guidance for identifying and resolving common issues that may occur during operation of the ECG monitor.

## 9.1 No Output / Flat Signal

- Verify that all electrodes are properly attached to the subject with good skin contact.
- Ensure adequate electrode gel is used and the skin is clean and dry.
- Confirm that the ECG monitor is powered ON.
- Check that both rechargeable Li-Po batteries are sufficiently charged.
- Inspect electrode cables and connectors for loose connections or damage.
- Ensure correct electrode placement according to the specified lead configuration.

## 9.2 Excessive Noise or Unstable Baseline

- Ensure electrode wires are not moving during measurement.
- Keep ECG cables away from mains power cables, transformers, and switching power supplies.
- Avoid operating the device near strong electromagnetic interference sources.
- Ensure the subject remains relaxed and minimizes body movement.
- Verify proper contact of the reference (ground) electrode.

## 9.3 Distorted or Attenuated R-Peaks

- Adjust the gain control potentiometer to an appropriate level.
- Verify correct operation of the amplifier gain stages.
- Ensure input filtering stages (high-pass and low-pass filters) are functioning correctly.
- Check for incorrect electrode placement or poor electrode contact.
- Avoid excessive gain settings that may cause amplifier saturation.

## 9.4 Signal Saturation or Clipping

- Reduce the gain using the gain control potentiometer.
- Ensure the ECG signal amplitude is within the designed input range.
- Check battery voltage levels, as low supply voltage may reduce amplifier headroom.
- Inspect the analog front-end for DC offset or drift.

## 9.5 Baseline Drift

- Allow the device to stabilize for a few seconds after power-on.
- Ensure stable electrode contact and minimal subject movement.
- Verify proper operation of the baseline stabilization (high-pass filtering) circuit.
- Avoid sudden posture changes during measurement.

## 9.6 Intermittent Signal Loss

- Check for loose or intermittent electrode lead connections.
- Inspect battery terminals and internal power connections.
- Ensure the rechargeable batteries are not deeply discharged.
- Verify that the gain control potentiometer is not at an unstable or extreme position.

## 9.7 Low Signal Amplitude

- Gradually increase the gain using the gain control potentiometer.
- Confirm correct electrode placement.
- Ensure electrode gel has not dried out.
- Verify that the batteries are fully charged.

## 9.8 Device Does Not Power On

- Recharge the Li-Po batteries using the specified charging method.
- Verify correct battery polarity and secure connections.
- Check the power switch for proper operation.
- Inspect the power circuitry for visible damage.

## 9.9 Charging Issues

- Use only the recommended charging voltage and current.
- Inspect the charging cable and connector for damage.
- Do not operate the device if batteries show signs of swelling or overheating.
- Allow sufficient charging time before use.

# 10.Maintenance

Proper maintenance of the ECG monitor ensures reliable operation, measurement accuracy, and extended device lifespan.

## 10.1 Storage

- Store the ECG monitor in a clean and dry environment.
- Avoid exposure to excessive humidity, dust, or direct sunlight.
- Store the device within the specified temperature range.

## 10.2 Cleaning

- Clean the electrodes after every use to prevent contamination.
- Use a soft cloth lightly moistened with a mild disinfectant or alcohol solution.
- Ensure electrodes are completely dry before the next use.
- Do not immerse the device or electrodes in water or cleaning liquids.

## 10.3 Battery Maintenance

- Recharge the Li-Po batteries using the recommended charging method.
- Avoid deep discharge of the batteries to prolong battery life.
- Do not use the device if the batteries show signs of swelling, leakage, or overheating.

## 10.4 General Precautions

- Do not attempt to disassemble the device.
- Inspect cables and connectors periodically for damage.
- If abnormal operation is observed, discontinue use and inspect the device.

# 11. Warranty

This ECG monitor is provided as part of an academic and experimental project. The device is intended for educational and laboratory use only.

The manufacturer provides a limited warranty covering manufacturing defects in materials and workmanship for a period of **six (6) months** from the date of assembly or first use.

This warranty does not cover:

- Damage caused by improper use, misuse, or unauthorized modification.
- Damage resulting from incorrect power supply, charging methods, or battery mishandling.
- Normal wear and tear of electrodes, cables, and connectors.
- Damage caused by exposure to moisture, liquids, or extreme environmental conditions.

The warranty is void if the device is disassembled or modified without authorization. The manufacturer shall not be held responsible for any indirect, incidental, or consequential damages arising from the use of this device.

**Note:** This device is not certified for clinical or medical diagnostic use.